

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **Digital Logic**

Code No: **CACS103** Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Define Digital Logic. Explain digital signal with its applications, advantages and disadvantages.
3. Define positional number system. Calculate following: a) Subtract 21 from 35 using 2's complement method b) Convert (62.75) into single precision floating point format
4. Define universal gate. Explain Universal gates with their graphical symbol, algebraic expression, truth table, and Venn diagram.
5. Define Decoder. Explain binary to octal converter with block diagram, truth table and logic diagram.
6. Simplify the Boolean function $F(w, x, y, z) = \pi(0, 2, 4, 6, 8, 10, 12, 14)$ and don't care conditions $d(w, x, y, z) = \pi(1, 3, 9, 11)$ using K-Map method for both SOP and POS form.
7. Define Multiplexer. Explain 4:1 multiplexer with its block diagram, truth table and logic diagram.
8. Write short Notes on (any Two): a) Parallel Adder b) PLAc) State Diagram

Group C

Attempt any TWO questions [2x10=20]

9. Explain JK and T Flip Flop with their Logic and Diagram, graphical symbol, characteristic table, characteristic equation and excitation table.
10. Differentiate between asynchronous and synchronous sequential circuits with example. Draw a block diagram, truth table and timing diagram to store 2001 in 4-bit SIPO register.
11. Define counter. Write a procedure to design a counter circuit. Design MOD-8 up counter.